PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	**Nizamaddeen Umar**
Program/Batch	B083 – NetworkWalks
Date	11 November 2026
Modules completed	W2-PM1 (Multiple Kali Linux Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. networkwalks.com (secured written permission already) 2. My own Windows PC and its network adapters
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems and devices where I had secured written permission, or on devices and systems that I own myself. All material in this report is for education and research purposes only. Do not use anything from it to break the law. The instructor, the author and NetworkWalks are not responsible for how this knowledge is used. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of employment and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report documents two practical activities from Week 2 of the Cybersecurity Programme at NetworkWalks (Batch B083): footprinting the networkwalks.com domain with multiple Kali Linux tools (W2-PM1) and scanning my own addresses with Zenmap (W2-PM5). The first covers the footprinting phase and the second covers scanning and network discovery, so together they show how an attacker can move from public information gathering to checking which hosts are live.

The footprinting commands were run in Kali Linux. The Zenmap scans were run on my Windows 10 PC. For each activity I describe what I did, what I observed and why it matters. The findings are reconnaissance observations. They are not confirmed vulnerabilities, and I did not attempt any exploitation.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for the reconnaissance and scanning activities.
WHOIS	Find domain registration details (registrar, dates, name servers).
WhatWeb	Fingerprint web technologies (web server, CMS, plugins, IP address).
Nslookup	Resolve the domain name to an IP address using DNS.
cURL (curl -I)	Read the HTTP response headers of the website.
Wafw00f	Detect whether a Web Application Firewall (WAF) protects the site.
DNSRecon	Enumerate DNS records (for example NS, MX, TXT/SPF and SRV).
Zenmap (Nmap GUI)	Run host-discovery (Ping Scan) and standard scans against my own addresses.
Windows CMD (ipconfig /all)	Identify local IP addresses, subnet masks, gateway and MAC addresses.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against networkwalks.com using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, cURL, Wafw00f and DNSRecon**. Each tool collects a different type of information about the target.

I used **WHOIS** to retrieve publicly available domain registration information. The registrar is GoDaddy.com, LLC and the name servers are NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM. The domain was created on 6 November 2019, last updated on 12 November 2025 and expires on 6 November 2027.

I used **WhatWeb** to identify the technologies behind the website. It identified Apache, WordPress and the WP Download Manager plugin.

Using **Nslookup**, I resolved networkwalks.com to **192.232.216.135**. The DNS server shown in the output was 202.113.16.41.

I used **cURL** with the -I option to read the HTTP response headers. The response was HTTP/2 200 from an Apache server, and it included a reference to the WordPress REST API endpoint /wp-json/.

I used **Wafw00f** to check for a Web Application Firewall. It identified **ModSecurity (SpiderLabs)**. This only shows that a WAF was detected. I did not run any attack-blocking test, so I cannot say how well it performs.

Finally, I used **DNSRecon** to enumerate DNS records. The output displayed DNS, mail and service-related records.

4.2 Network Scanning with Zenmap

For the second activity I used Zenmap to run host discovery from my own Windows PC. I first used ipconfig /all to record the details of my two network adapters. They belong to two different networks, so I list them separately.

Wi-Fi adapter (active LAN)	Value
PC hostname	DESKTOP-QCLTH13
Operating system	Windows 10 (version 10.0.19045.6466)
Adapter	Intel(R) Dual Band Wireless-N 7260
Local IP address	10.22.190.101
Subnet mask	255.255.192.0
LAN subnet	10.22.128.0/18
Default gateway	10.22.128.1
MAC address	A0-A8-CD-A5-00-AA

VirtualBox Host-Only adapter	Value
Adapter	VirtualBox Host-Only Ethernet Adapter
IP address	192.168.56.1
Subnet	192.168.56.0/24
MAC address	0A-00-27-00-00-14

The MAC addresses above come from the Windows ipconfig /all output. They are not taken from the Zenmap scan output.

I then ran Zenmap scans against 10.22.190.101, my own Wi-Fi address. I also ran a Ping Scan against 192.168.56.1, my VirtualBox Host-Only address, as shown in the Zenmap screenshot.

Scan	Target	Result
Zenmap scan 1	10.22.190.101	Host is up. 1 IP address (1 host up) scanned in 0.82 seconds.
Zenmap scan 2	10.22.190.101	Host is up. 1 IP address (1 host up) scanned in 0.85 seconds.
Zenmap Ping Scan	192.168.56.1	One host reported up (shown in the Zenmap screenshot).

What these results show: Zenmap confirmed that 10.22.190.101 and 192.168.56.1 are up. These are addresses on my own PC, and they belong to different networks (10.22.128.0/18 and 192.168.56.0/24).

Limitations: I did not scan the full Wi-Fi subnet, so I have not established how many other hosts are live on 10.22.128.0/18. I discovered no other IP or MAC addresses and I make no claim about other devices. I also have not included a Zenmap topology PDF as evidence.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and scanning activities, I identified the following potential risks. These are observations from reconnaissance and are not confirmed vulnerabilities.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Website technology information exposed	WhatWeb identified Apache, WordPress and WP Download Manager.	Attackers may use exposed technology and version details to look for known issues; not a confirmed vulnerability.	● **Medium**
2	Server IP address identifiable	Nslookup resolved networkwalks.com to 192.232.216.135.	Shows the network location of the web server. Resolving a public domain is normal.	● **Low**
3	HTTP technical information exposed	cURL returned HTTP/2 200, an Apache server header and a /wp-json/ (WordPress REST API) reference.	May assist fingerprinting and further enumeration of the web service.	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs).	Reveals information about the web application's security layer. No attack-blocking test was performed, so WAF effectiveness is not assessed.	● **Low**
5	DNS infrastructure information exposed	WHOIS showed HostGator name servers; DNSRecon displayed DNS, mail and service-related records.	Public DNS data can help build a picture of the hosting and mail infrastructure.	● **Low**
6	Network discovery coverage incomplete	Zenmap confirmed 10.22.190.101 (Wi-Fi) and 192.168.56.1 (Host-Only) as up. The Wi-Fi subnet 10.22.128.0/18 was not scanned in full.	The number of live hosts on the Wi-Fi subnet is unknown, so unknown devices cannot be ruled in or out.	● **Info**

Risk level key: ● Medium ● Low ● Info (no confirmed risk)

The exercises involved only information gathering and host discovery. I did not attempt exploitation or vulnerability validation. The presence of information such as a software version, an IP address or a DNS record does not mean the system is vulnerable. Further authorised testing would be needed to confirm any actual weakness.

6. Recommendations

Based on these observations, I recommend the following improvements:

1. **Review publicly exposed technology information**

   Check what the website reveals about its web server, CMS and plugins, and remove anything that is not needed.

2. **Keep software updated**

   Keep WordPress, plugins (including WP Download Manager), themes and Apache patched and review vendor security advisories.

3. **Review HTTP response headers**

   Review headers and the /wp-json/ reference to decide whether unnecessary technical information should be hidden or restricted.

4. **Review DNS records regularly**

   Check DNS records periodically and remove outdated or unused entries. Confirm that SPF and other mail-related records are correct.

5. **Configure and monitor the WAF**

   Keep ModSecurity rules up to date, review its logs, and test its configuration only with written authorisation.

6. **Perform regular internal network discovery**

   Scan the full Wi-Fi subnet (10.22.128.0/18) with permission, and repeat the scan regularly to build a baseline of active hosts.

7. **Investigate unknown devices**

   Compare discovered IP and MAC addresses with a known device list and investigate any device that cannot be identified.

8. **Maintain network documentation**

   Record each network separately (the Wi-Fi LAN and the VirtualBox Host-Only network) with IP ranges, gateways and known devices.

9. **Perform security testing with authorisation**

   Only scan or test systems I own or have written permission to test, and keep a record of scope and permission.

7. Conclusion

During Week 2 of the Cybersecurity Programme at NetworkWalks I completed practical footprinting and network scanning exercises.

In the footprinting activity, WHOIS gave domain registration details, WhatWeb identified Apache, WordPress and a download-manager plugin, Nslookup resolved the domain to 192.232.216.135, cURL showed the HTTP headers and a WordPress REST API reference, Wafw00f detected ModSecurity, and DNSRecon displayed DNS records.

In the scanning activity, Zenmap confirmed that my Wi-Fi address (10.22.190.101) and my VirtualBox Host-Only address (192.168.56.1) were up. These are on separate networks, and I have not yet determined how many hosts are live across the Wi-Fi subnet.

I learned that public information can reveal a great deal about a target, and that findings from reconnaissance must be reported carefully and kept separate from confirmed vulnerabilities. All activities were carried out within an authorised scope.

8. Evidence Collected

The figures below are the evidence for each activity, in the order they were performed.

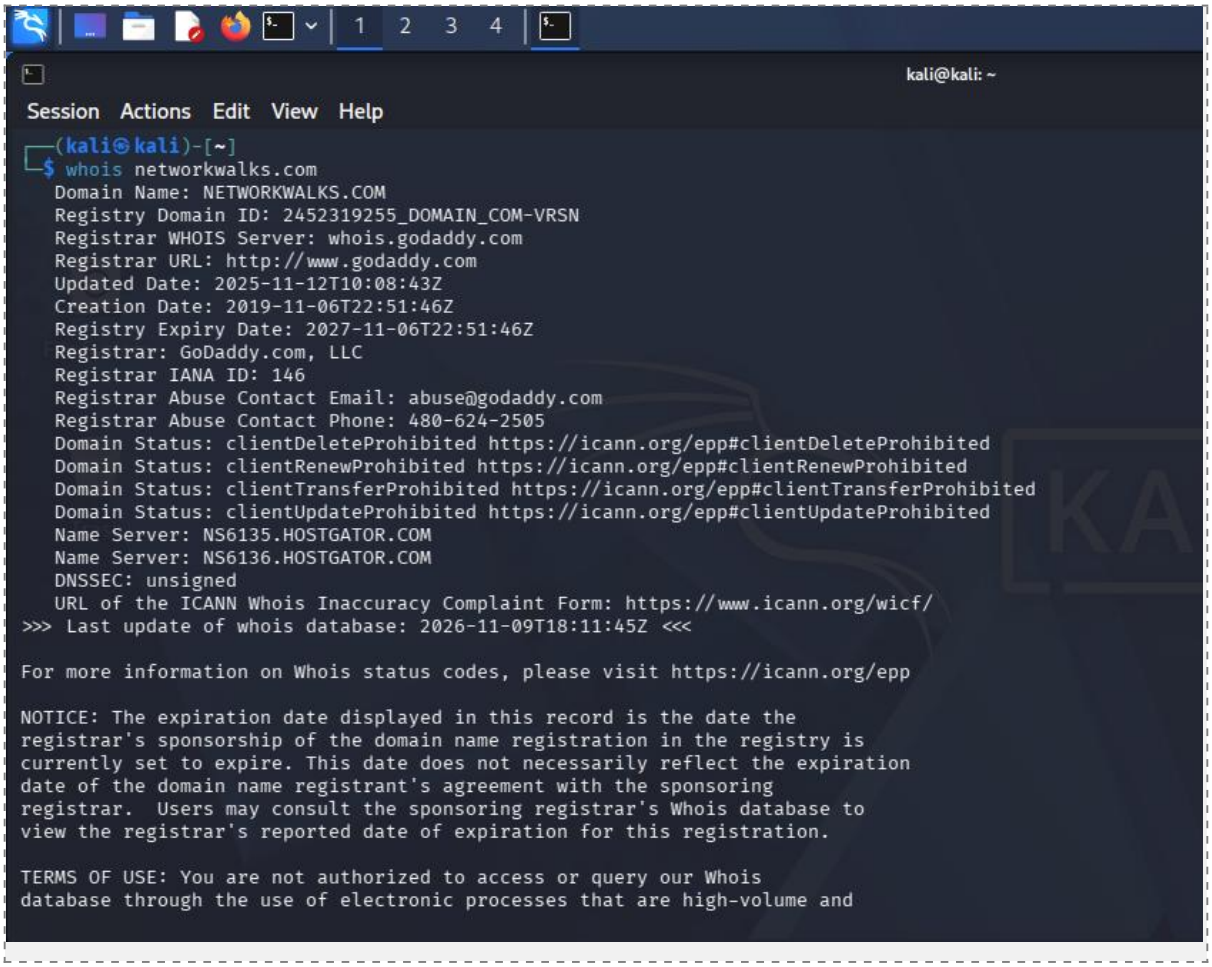


Figure 1: WHOIS output for networkwalks.com

Registrar, name servers and creation, updated and expiry dates.

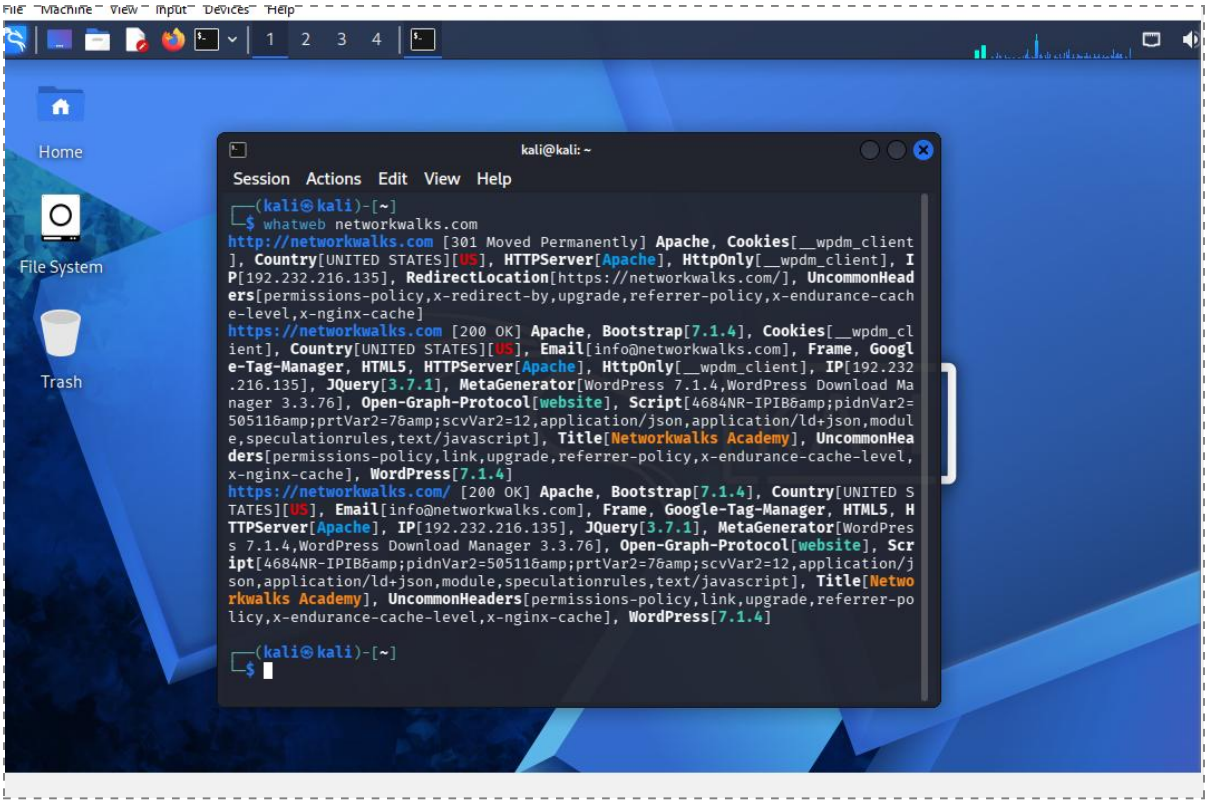


Figure 2: WhatWeb output for networkwalks.com

Web server, CMS and plugin identification.

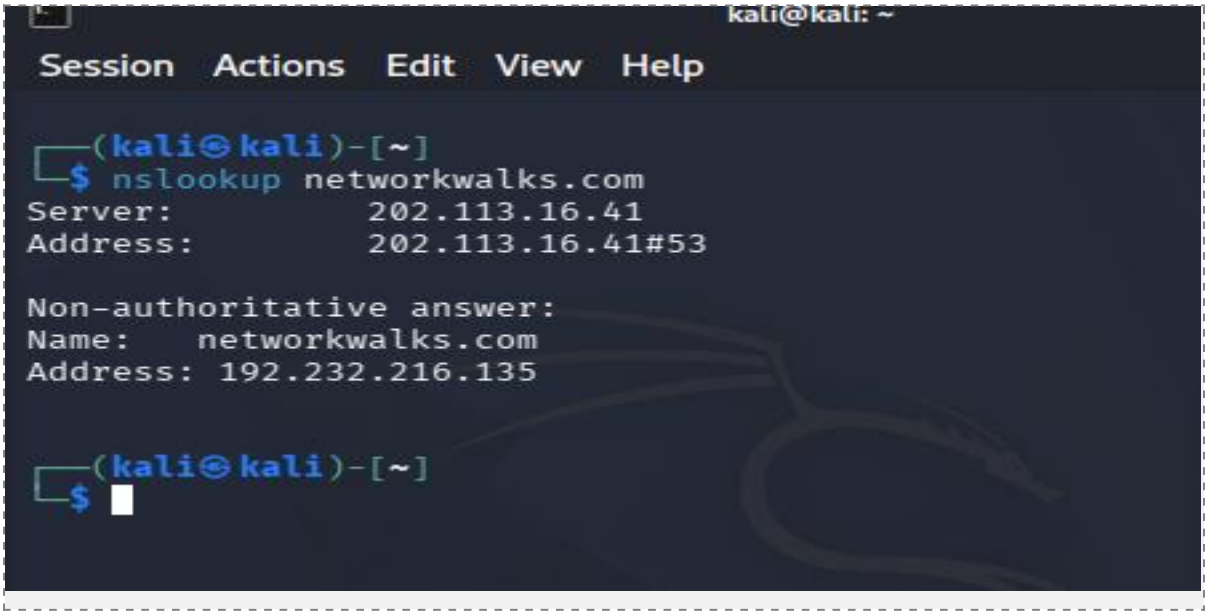


Figure 3: Nslookup output for networkwalks.com

Resolved IP address and DNS server used.

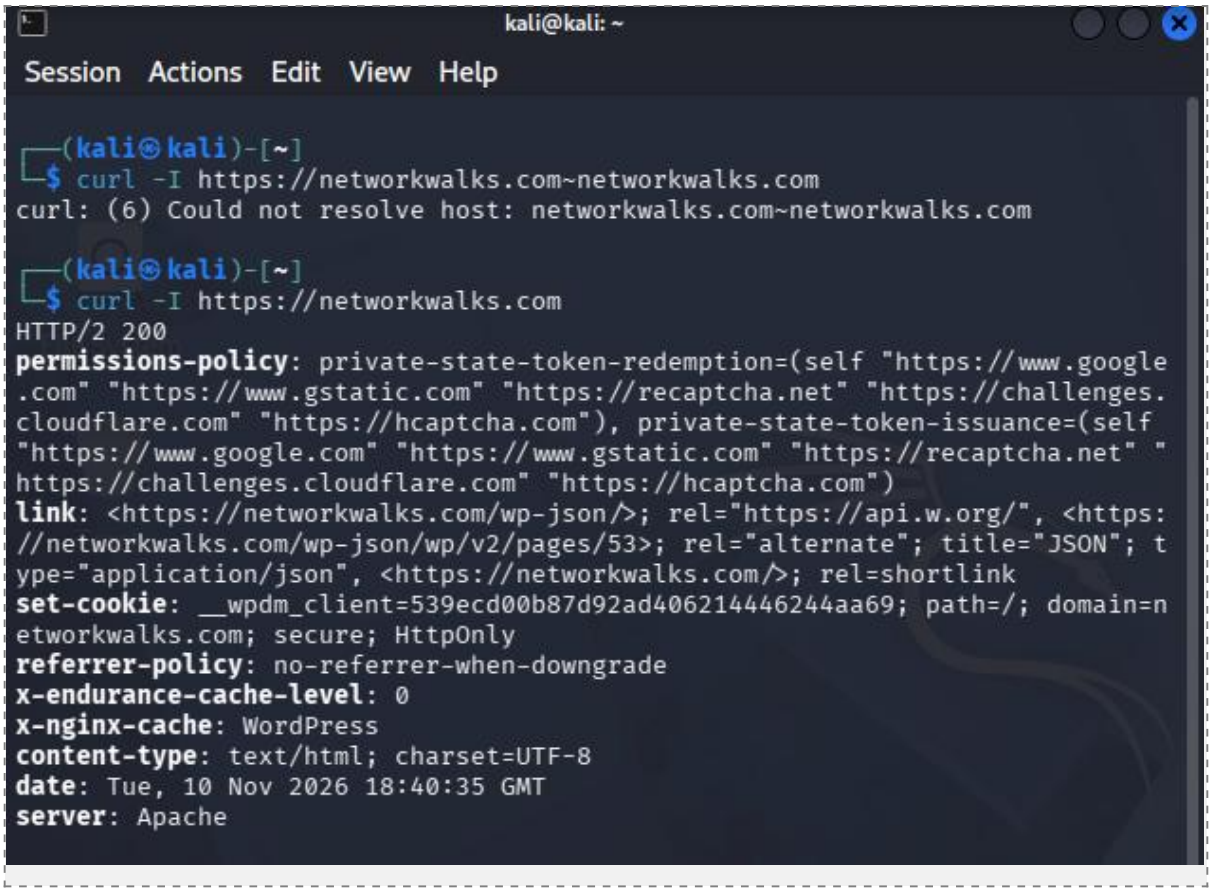


Figure 4: cURL (curl -I) HTTP response headers

HTTP status, server header and /wp-json/ reference.

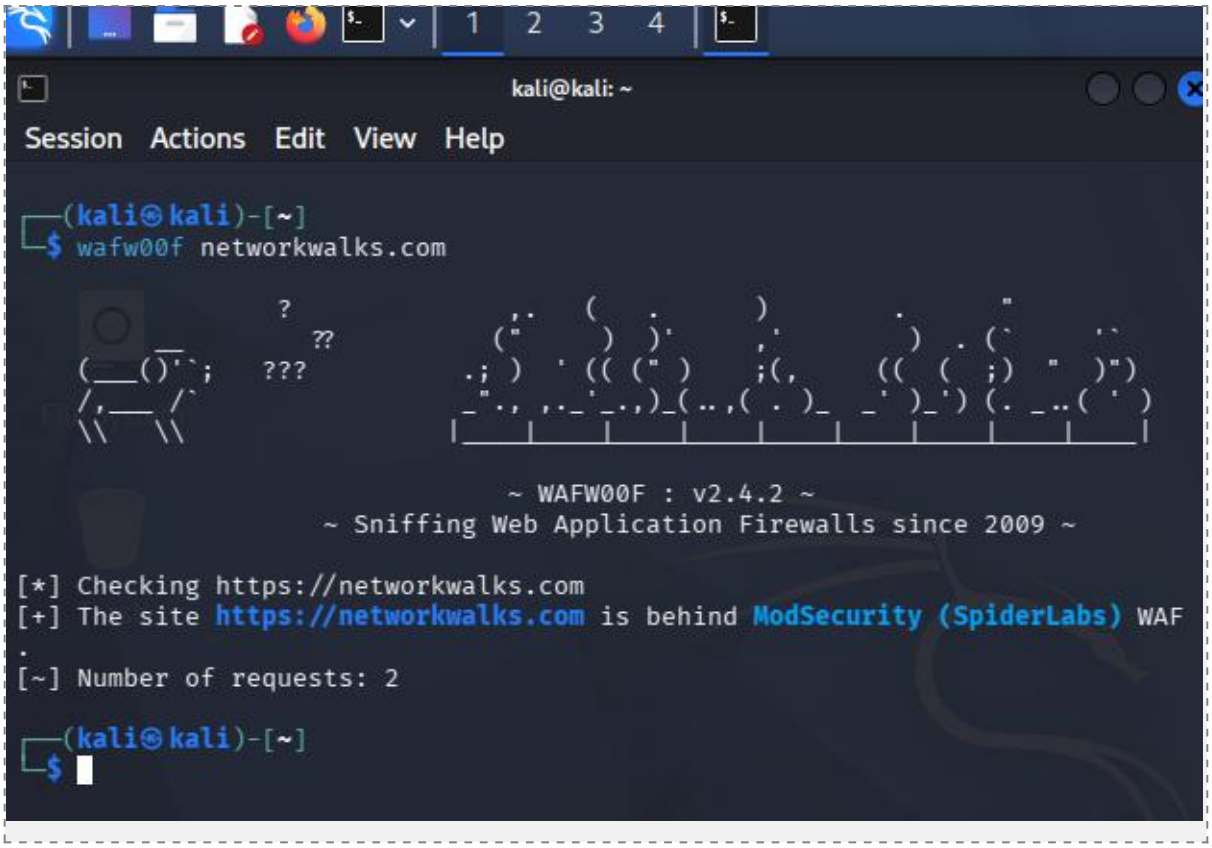


Figure 5: Wafw00f detection result

WAF detection result.

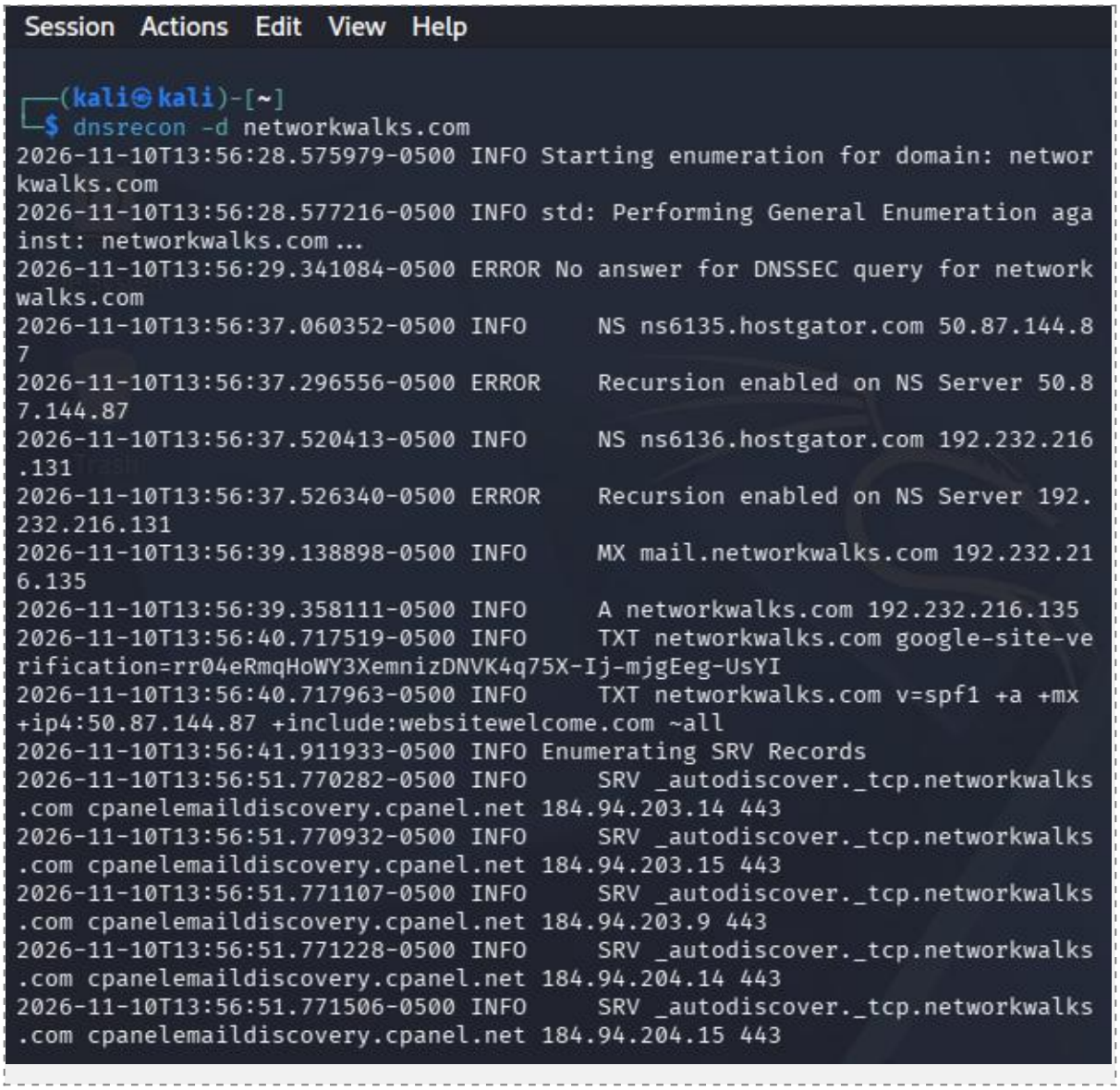


Figure 6: DNSRecon DNS enumeration output

DNS, mail and service-related records.

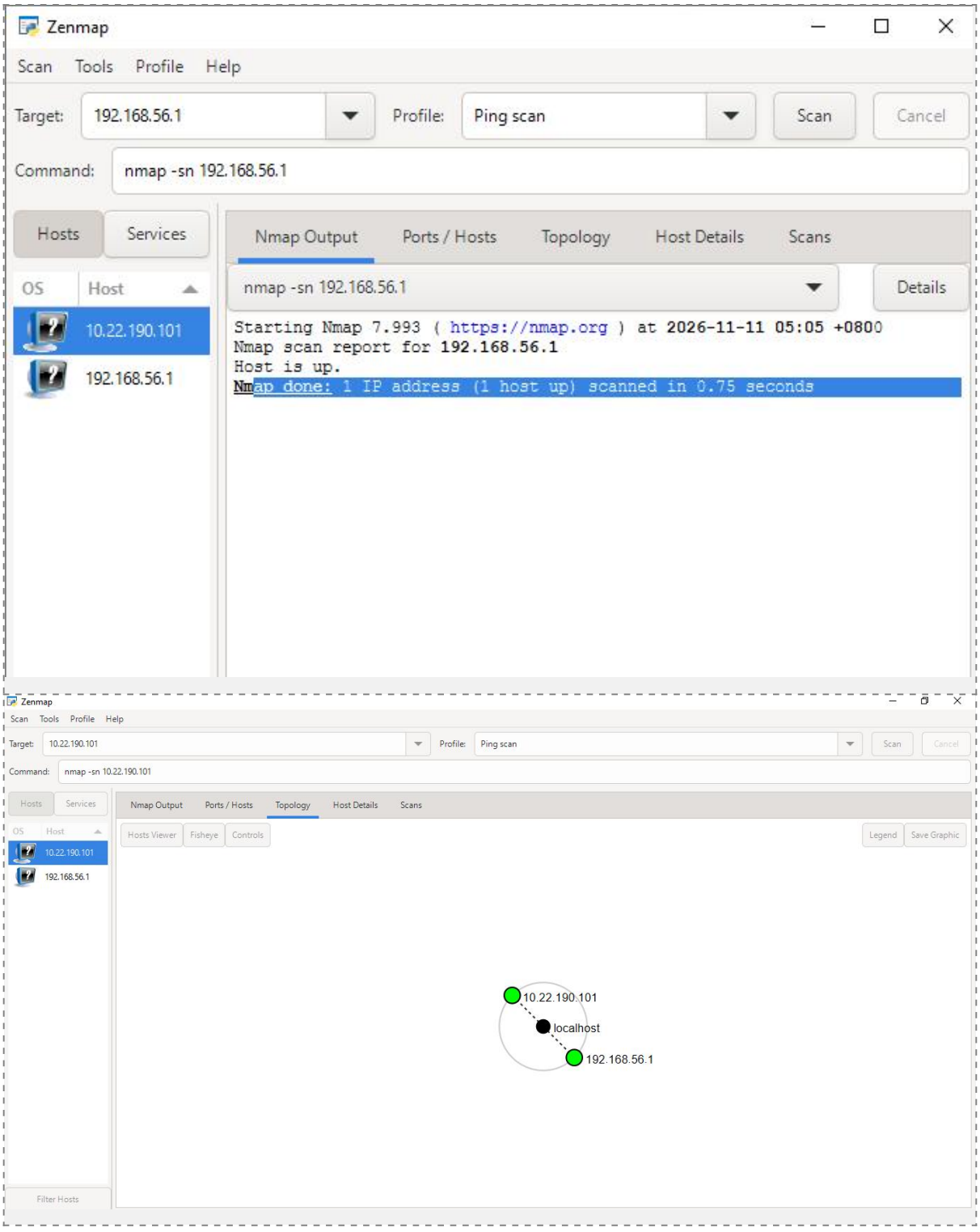


Figure 7: Zenmap scans of 10.22.190.101 and 192.168.56.1 and Topology

Host-up results for the Wi-Fi address and the Ping Scan of the Host-Only address.

Author

Nizamaddeen Umar

Cybersecurity Professional, Batch B083 – NetworkWalks

Github: NIZAMADDEEN/NETWORKWALKS-BO83-WK2-PM1-CYBERSECURITY-PROJECT1-FOOTPRINTING

LinkedIn: https://www.linkedin.com/in/nizamaddeen-umar-886064145/

📌 Project Information

Program Name: Cybersecurity Programme at NetworkWalks | **Week:** 02 | **Project:** W2-PM-FINAL